

TITLE: NeuroResuscitation in VP and VA ECMO

PROTOCOL: Non-invasive neuromonitoring in Extracorporeal Membrane Oxygenation (ECMO) patients, with intervention guidance

PURPOSE: The Extracorporeal Life Support Organization (ELSO) has recently released guidance suggesting standardized neuromonitoring in ECMO patients; however, evidence and literature describing how exactly to monitor patients is lacking, and how neuromonitoring changes management is unclear.¹ Neurologic complications in ECMO independently predict poor outcomes, and addressing abnormal neurophysiology may have a role in the prevention and/or early identification of acute ECMO-related brain injury, with management implications.²

The purpose of the ECMO neuromonitoring protocol is threefold. Firstly, to identify patients at risk for neurologic complications with a multimodal non-invasive neuromonitoring approach. Second, to provide a diagnostic algorithm for patients with abnormal neuromonitoring parameters and guide providers on when to obtain neuroimaging on select patients. And third, to outline a strategy of intervention to optimize cerebral perfusion in patients with abnormal neuromonitoring, neuroimaging, and/or laboratory diagnostic results.

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DEFINITIONS:

- A. **ECMO configuration nomenclature** – Per the ELSO Maastricht Treaty³ for Extracorporeal Life Support (ECLS) nomenclature, standardization of ECMO configuration abbreviations is imperative. The membrane lung is represented by a hyphen (“-”). Locations from which deoxygenated blood is drained are on the left of the membrane lung, and locations to which oxygenated blood is returned are on the right of the membrane lung. Major drainage and return sites are represented by capitalized letters, where minor sites are represented by lowercase letters. Further nuances of nomenclature are outside the scope of this document. The following are examples of commonly utilized ECMO configurations:
- a. **V-A ECMO** – a configuration where deoxygenated blood is drained from a vein (i.e. inferior vena cava), and oxygenated blood is returned to an artery. (I.e. femoral artery)
 - b. **V-AV ECMO** – a configuration where deoxygenated blood is drained from a vein, and oxygenated blood is returned to both an artery AND a vein.
 - c. **V-P ECMO** – a configuration where deoxygenated blood is drained from a vein, and oxygenated blood is returned to the pulmonary artery
- B. **Patient selection** – This protocol is primarily intended for use in patients in whom blood is being returned to the arterial circulation. Examples of this are in V-AV ECMO, V-A ECMO, and in more rare configurations such as LAV-A ECMO (Left atrial drainage and arterial return). However, this list is not all-inclusive, and the neuromonitoring pathway is reasonable to institute on any ECMO patient deemed appropriate by the cannulating team. However, the majority of patients with return of blood flow to the venous side (i.e., V-V ECMO or V-P ECMO) will not require intensive non-invasive neuromonitoring as outlined in this protocol.
- C. **Mixing Point** - The location within the aorta where anterograde blood flow from the left ventricle meets retrograde reinfused blood flow from the ECMO circuit.
- a. **Antegrade Blood Flow** – the direction of blood flow from the heart
 - i. **Native Lung Zone** – the region of blood flow extending from the aortic valve to the mixing point
 - b. **Retrograde Blood Flow** – the direction of blood flow from a peripheral ECMO circuit in which blood is returned to a peripheral artery (excluding axillary arteries). This blood flow is in the opposite direction from native blood flow
 - i. **Membrane Lung Zone** – the region of blood flow returning distally from an arterial return cannula, extending to the suspected mixing point
- D. **Differential gas exchange**⁴
- a. **Differential oxygenation**

- i. Without hypoxemia or hyperoxemia – when PaO₂ differs between membrane lung and native lung zones, but without hypoxemia or hyperoxemia
- ii. With regional hypoxemia – when PaO₂ is < 70 in either the membrane lung or native lung zones
 - 1. can be with or without evidence of tissue ischemia (i.e., lactate elevation)
- iii. With regional Hyperoxemia – when PaO₂ > 150 in either the membrane lung or native lung zones

b. Differential carbon dioxide

- i. With normocapnia – when PaCO₂ is different, but within normal range in both the membrane and native lung zones
 - 1. **Normocapnia** – a PaCO₂ 35-45 mmHg
 - 2. **Goal PaCO₂** – may differ from normocapnia on a patient-by-patient basis. Generally speaking, after initial sweep adjustments as outlined below, the ultimate goal PaCO₂ post cardiac arrest will be between 40-45 mmHg. This goal should be achieved *gradually* (see below)
- ii. With regional hypocapnia – when there is hypocapnia in either the membrane or native lung zones
- iii. With regional hypercapnia – when there is hypercapnia in either the membrane or native lung zones

- E. Pulsatility** – as opposed to non-pulsatility, pulsatility is defined as when the brain is receiving pulsatile blood flow. This is formally defined as pulsatility noted on transcranial doppler exams. However, recognizing that transcranial doppler is not a readily available diagnostic, particularly in crisis, pulsatility is also defined as a LUE pulse pressure > 15
- a. While a low pulse pressure may also predict poor cardiac recovery, a pulse pressure <10 has been associated with the development of neurologic complications.⁵

- F. ECMO Associated Brain Injury**– commonly includes hypoxic ischemic brain injury, ischemic stroke, hemorrhagic stroke (IPH), subdural hematoma, cerebral venous sinus thrombosis

G. Neurologic exam

- a. For the sake of this protocol, a patient is considered “**awake**” if they are *clearly* following commands. Notably, patients typically follow central simple commands prior to peripheral commands. Thus, a patient who appears to be squeezing the

provider's fingers, but is not opening eyes or extruding tongue, is not considered to be "clearly following commands."

b. Reliable Motor Exam

- i. Strength score – a scoring system to grade a patient's unilateral extremity strength
 - a. 0/5 – no movement of contraction
 - b. 1/5 – flicker or trace contraction (Activating)
 - c. 2/5 – movement is possible with gravity eliminated
 - d. 3/5 – antigravity movement, but not against added resistance
 - e. 4/5 – movement occurs against some applied resistance
 - f. 5/5 full power against any resistance
2. For the sake of this protocol, the presence of a reliable motor exam requires at least 3/5 strength in all extremities. Patients without 3/5 strength in even one extremity are not considered to have a "reliable motor exam."

- H. **Acute Brain Injury** – Also known as "primary brain injury." The immediate damage the brain sustains after a stimulus or insult which is known to result in brain injury. In post-cardiac arrest, this can be considered the injury as a result of the initial low/no flow hypoperfusion state, and is referred to as Hypoxic Ischemic Brain Injury (HIBI)
- I. **Secondary Brain Injury** – A delayed, progressive brain injury that occurs as a result of the brain's poorly adaptive response to an acute brain injury. Contributing factors include inflammatory changes, cerebral edema, loss of cerebral autoregulation, and development of reactive oxygen species. Importantly, there are many clinical targets that, when addressed, may prevent or significantly mitigate the development of secondary brain injury. Relevant targets for optimization may include oxygen delivery, cerebral metabolic demand, and several cerebral perfusion parameters, all of which are addressed in the body of this protocol.
- J. **Coma** – a state characterized by the absence of both arousal and awareness of one's environment. Usually, prerequisite is a RASS score of -5
- K. **Neuromonitoring Event** – See below, under the heading "EVENT PROTOCOLS". Separated into two events: global hypoperfusion crisis and regional hypoperfusion crisis.
- L. **Self-fulfilling Prophecy Bias** – a clinical bias in which, in patients where a neurologic outcome is assumed to be poor, care is altered in a way that contributes to the development of this poor prognosis.

NEUROMONITORING MODALITIES

A. Near-Infrared Spectroscopy (NIRS) *

- a. RSO₂ – regional cerebral oxygen saturation. This is a surrogate for cerebral blood flow, as measured by a spectroscopy method
 - i. An RSO₂ > 55-60 is considered normal
 - ii. An RSO₂ < 40 has been associated with acute brain injury in ECMO patients^{6,7}

B. Transcranial Doppler Ultrasound (TCDs)

- a. Peak Systolic Velocity (PSV)
- b. End Diastolic Velocity (EDV)
- c. Mean Flow Velocity (MFV) – an indicator of cerebral blood flow in the measured cerebral artery
 - i. Calculation: $(PSV + 2 \times EDV) / 3$
 - ii. Normal values are considered to be < 80-120 cm/s. Healthy controls are expected to have average MFV of 60 cm/s
- d. Pulsatility Index (PI)
 - i. Calculation – Gosling's Formula: $(PSV - EDV) / MFV$
 - ii. Normal range is considered ~ 0.5 – 1.2
 - 1. Elevated values typically result from increased distal cerebral vascular resistance, and may indicate developing neuro-pathology such as cerebral edema
 - 2. Decreased values, in this patient population, may indicate compromised cerebral blood flow and/or compromised native heart ejection
 - iii. Absence of a recordable PI has been independently associated with the development of Intracerebral Hemorrhage (ICH) in ECMO patients⁸
- e. Resistive Index (RI)
 - i. Calculation: $(PSV - EDV) / PSV$
 - 1. RI < 0.75 is considered normal
 - ii. Elevated RI is more specific, but less sensitive for the development of cerebral edema, compared to PI
- f. Microembolic events (MES)⁹
 - i. MES are associated with decreased LV ejection fraction, use of LV unloading devices (i.e. percutaneous LVAD), and VA compared to VV ECMO.
 - ii. Frequency of MES events have not clearly been associated with ECMO-related brain injury

C. Pupillometry

- a. Neurologic Pupil Index (NPI) – as measured by an infrared pupillometer. Abnormal NPI measurements are strongly associated with poor neurologic outcomes, however improvement of NPI has been associated with good neurologic outcomes in ECMO patients.¹⁰
 - i. Values < 3 is considered abnormal
 - ii. Asymmetry > 0.7 is considered abnormal

CORE PROCEDURE:

OBJECTIVE

- This protocol has two main objectives:
 1. Identify patients both with acute brain injury and at a *modifiable* risk of secondary brain injury. Inherent in this is identifying patients at risk of clinically important differential gas exchange.
 2. Guide interventions designed to correct modifiable abnormalities, with the ultimate goal of reducing the risk of development of secondary brain injury with the designed interventions. While data is sparse, evidence suggests that when interventions to improve neurologic monitoring parameters are successful, the development of secondary brain injury can be mitigated¹¹
- It is important to note that in many patients, modifications and interventions performed may not ultimately improve the outcome. However, it is more important to note that, in the light of self-fulfilling prophecy bias, altering management based on a presumed poor prognosis (especially when prematurely determined) may contribute to the development of that poor prognosis. Thus, it is strongly recommended that this protocol be utilized in its entirety in all eligible patients, regardless of presumed prognosis, during the duration of ECMO support

PATIENT SELECTION

- This protocol is intended for use in all patients with an arterial return cannula. However, it is also available for use in any ECMO patient of any configuration, at the discretion of either the cannulating team or the critical care team.

- As of the date of this protocol's publication, this protocol is primarily intended for use in patients admitted to the Thoracic ICU (87)

NURSING RESPONSIBILITIES

- A. Upon arrival, ensure bilateral NIRS is set up, and monitoring is at bedside
 - a. Notify the critical care medicine or ERS team for any of the following
 - i. NIRS < 60% in any cerebral lead
 - ii. Asymmetry > 10% in any cerebral lead
- B. Perform NPI during initial exam, and then q1h until deescalated by critical team
 - a. Notify critical care medicine or ERS team, for any of the following
 - i. NEW NPI < 3 in any pupil
 - ii. NEW Asymmetry > 0.7
- C. Monitor arterial line blood pressures per typical ICU nursing protocol
 - a. In addition to standard management, notify critical care medicine or the ERS team for any of the following
 - i. Pulse pressure < 15 in any arterial waveform
 - ii. Loss of pulsatility in any arterial waveform

STANDARD OPERATING PROCEDURES

- A. **Targeted Temperature Management (TTM)** – TTM may be indicated in patients post-cardiac arrest, cannulated for VA ECMO. With the publication of the TTM2, the utility of temperature control post-cardiac arrest has recently been called into question. However, the therapy has never been studied in a randomized fashion in the ECMO patient population. There is evidence that patients with worse brain injury, based on neurologic exam (when excluding patients with cerebral edema on index head CT) and/or more severe organ dysfunction and shock, as quantified by higher values of the respiratory and cardiac components of the SOFA score, had higher survival with TTM to 33 °C.¹² The latter criterion is likely to be met in all ECMO patients, by definition. Furthermore, two recent seminal studies of ECPR both used TTM to 33 °C in their study protocols, which likely embodies the strongest evidence to guide temperature control in ECPR patients.^{13,14} Thus, the following protocol should be followed for ECPR or patients on VA ECMO who have suffered a pre-cannulation cardiac arrest:

- a. For patients who are either following commands or have a purposeful neurological exam, TTM to 37.5 °C should be initiated
 - b. For patients who have a non-purposeful neurological exam, AND loss of at least one brainstem reflex, TTM to 33 °C should be initiated
 - c. For patients who have a non-purposeful neurological exam, AND either refractory shock or ARDS, quantitatively defined as a combined respiratory and cardiac SOFA score > 4, TTM to 33 °C should be initiated
 - d. For patients who have a non-purposeful neurologic exam and CT findings with cerebral edema, TTM to 33 °C should be initiated. However, it is notable that the clinical effect in this population is questionable
 - e. For patients with a purposeful neurological exam, without the loss of brain stem reflexes, and either refractory shock or ARDS, quantitatively defined as a combined respiratory and cardiac SOFA score > 4, TTM to either 33 °C or 37.5 °C is reasonable
 - f. Goal temperature should be maintained for 24 hours, followed by a controlled rewarming of 0.25-0.5 °C/hr, until 37.5°C is reached. The temperature of 37.5 °C should then be maintained for at least 72 hours.
 - g. The cannulating team may rarely adjust the temperature goal based on clinical factors, on the initial assessment at the time of cannulation.
 - i. However, temperature should be maintained for the entire duration of 24 hours, and it should be an extremely rare event that the patient is rewarmed prior to the completion of the protocol. Bradycardia, electrolyte disturbances, and abnormal coagulation studies are not considered indications for rewarming in the post-cardiac arrest literature. Active rewarming in the first 24 hours post-arrest should be highly avoided
- B. Biomarkers of Brain Injury** – multiple biomarkers have been proposed to use as a tool to aid in neuroprognostication in cardiac arrest. At the current time, the most studied biomarker is neuron-specific enolase
- a. **Neuron-specific Enolase (NSE)** – the biomarker with the most evidentiary support is NSE, and elevated NSE drawn within 72 hours has been demonstrated to be predictive of poor outcome. However, it is recognized that NSE may be confounded by the ECMO circuit, by the presence of hemolysis, or by other factors not yet recognized, and supporting data is lacking in the ECPR patient population¹⁵.
 - i. NSE is to be drawn with index blood work, as well as daily with morning labs for a total of 5 draws. Use in neuroprognostication should only be done in a multimodal format, with credence to the assay's potential limitations.
- C. Line placement** – All VA ECMO patients should receive the following lines initially

- a. **Pulmonary Arterial Catheter (PAC)** – literature currently supports the use of invasive cardiac monitoring via PAC in cardiogenic or mixed shock, particularly in mechanical circulatory support.¹⁶ It is worth noting that previous works that demonstrated a lack of benefit specifically excluded this patient population.
 - b. **Rt upper extremity Arterial Line**
 - i. Preferred site is radial
 - c. **Lt Radial Arterial Line**
 - i. Preferred site is radial
 - ii. If unable to place an arterial line into the Lt Radial artery successfully, please discuss next steps with the ERS team at shift change/sign out. Under certain circumstances, a more proximal LUE arterial line may be warranted
- D. **Neurologic examination** – All patients should have an hourly sedation hold for a comprehensive neurologic exam hourly, until exam frequency is de-escalated by the rounding teams. The following exceptions can be considered
- a. TTM to 33C – sedation is required in these patients. Furthermore, shivering (which is common and predictive of good neurologic outcomes) must be treated aggressively to lower the Cerebral Metabolic Rate of Oxygen (CMRO₂). Treatment options are standardized and can be found in the TTM protocol and order set
 - b. Profound hemodynamic instability – if sedation holds/re-sedation are contributing to profound hemodynamic instability, neurological exams can be de-escalated in frequency or temporarily deferred
 - c. Acute lung injury – in patients without an ECMO configuration that supports lung injury (i.e., a venous side return cannula), lung injury may require continuous sedation
 - i. On the contrary, in patients with ECMO configurations that offer pulmonary support, such as V-V, V-AV, or V-P ECMO, sedation should be weaned with the target of extubation of the patient, if appropriate
 - d. Concern for severe and/or ongoing brain injury and/or cerebral edema – in these patients, lowering the CMRO₂ via continuous sedation is important to mitigate the development of secondary brain injury
- E. **Noninvasive monitoring**
- a. **NIRS** – all patients should have the following NIRS monitors upon admission to the ICU. NIRS modules have the capability for all four leads to be attached simultaneously
 - i. Rt frontal
 - ii. Lt frontal
 - iii. Rt lower extremity

- iv. Lt lower extremity
- v. In rare circumstances where all 4 leads are not available, the lower extremity that does NOT have an arterial return cannula can be deferred from NIRS attachment. Under no circumstances should either of the cerebral leads be deferred
- b. **EEG** – should be ordered as continuous, and be attached as soon as a monitor becomes available. Attachment should not be deferred in anticipation of possible future imaging studies.
 - i. Do not order spot (<24 hrs) EEG – it is rare that this study is of use in this patient population
 - ii. EEG discontinuation is at the discretion of the ERS and neurocritical care teams, in conjunction with the rounding team

****ROUTINE NEUROIMAGING****

- A. Background – unlike typical ICU patients, head CT is extraordinarily high yield in patients on VA ECMO, including in patients without focal neurologic deficits. Head CTs obtained for reasons other than neurologic deterioration can have a yield as high as 17-20%¹⁷
 - a. There are limitations of neuroimaging on VA ECMO patients (See appendix B). Specifically, in patients with differential gas exchange and mixing points in the aortic arch, CT perfusion can have a false positive result. It is highly important to have a multidisciplinary discussion between stroke neurology, ERS, and CCM, particularly prior to consideration of stroke-specific therapy, such as thrombolysis and/or mechanical thrombectomy
- B. Procedure
 - a. All ED patients on ECMO will receive a non-contrast head CT, as a component of a full body CT scan, prior to arrival to the ICU
 - i. If the patient is already in the ICU, or arrives at the ICU without neuroimaging, a stat non-contrast head CT will be obtained as soon as feasible
 - b. All patients on VA ECMO will receive a non-contrast head CT at 24 hours
 - c. All patients on VA ECMO will receive a non-contrast head CT at 72 hours, unless all of the following criteria are met
 - i. Patient is not comatose
 - ii. Patient is reliably following central and peripheral commands
 - iii. Patient has a reliable and non-focal neurologic exam

- iv. The patient has not had any neuromonitoring-triggered events as listed below

IMMEDIATE POST-CANNULATION PROCEDURES

- A. Prior to ICU arrival/immediately upon arrival to ICU – these are considered immediate, “tier 0” interventions. Note that, depending on the availability of ED resources, some or all of these procedures may be deferred until arrival at the ICU; however, in an ideal situation, they have occurred prior to ICU arrival. Also note that any of the following may be elected to be deferred by the cannulating team, for the sake of timeliness. A common reason for deferring until ICU arrival would be not to delay cardiac catheterization/PCI
 - a. Standard Operating Procedures* - See above
 - b. Routine Imaging Studies** - See above
 - c. Baseline blood gas collection
 - i. Check bilateral radial arterial blood gases to establish baseline PaO₂ and PaCO₂
 - ii. Note pulse pressure of bilateral arterial lines. Pulse pressure < 20 is considered abnormal
 - d. Sweep adjustment
 - i. Prior to ABG results – set sweep gas flow at 1LPM
 - a. Note that sweep gas flow should never fall below 0.5LPM in VA ECMO
 - ii. After obtaining ABG results, adjustments will be made based on LUE pulse pressure
 - 1. If LUE pulse pressure < 15, primarily adjust sweep gas flow to target goal LUE PaCO₂
 - a. Adjust ventilator based on available clinical parameters, however blood gas measurements may not be reliable
 - 2. If LUE pulse pressure > 15, adjust sweep gas flow AND ventilator parameters to target goal PaCO₂
 - e. Membrane FSO₂ adjustment
 - i. Early PaO₂ > 150mm Hg post-ECMO initiation has been associated with the development of secondary brain injury¹⁸

- ii. Attach gas blender to sweep gas as soon as possible – ideally, this is in the emergency department prior to transfer, but this should not delay transfer to the cardiac cath lab. Attachment, and as implicit, titration of F_{SO2} is imperative and should be initiated prior to ICU transfer.
 - 1. If LUE pulse pressure < 15, primarily adjust F_{SO2} to target goal LUE PAO₂ or oxygen saturation
 - a. Adjust ventilator to target goal RUE PAO₂ or oxygen saturation
 - 2. If LUE pulse pressure > 15, adjust F_{SO2} AND ventilator to target goal RUE PAO₂ or oxygen saturation
- f. Goal blood gas parameters
 - i. PaCO₂ – target all of the following
 - 1. If initial PaCO₂ is > 50mmHG, adjust parameters slowly, targeting no greater than a 50% decrease in the first 24 hours
 - 2. Goal PaCO₂ 40-50 mmHg
 - ii. PaO₂ – goal 80-140 mmHg
 - iii. Oxygen saturation – goal 92%-96%
 - 1. Oxygenation goals can be adjusted in patients with a highly suspected history of chronic hypoxemia, on a rare and case-by-case basis
 - a. Should never be adjusted in patients without this suspected history
 - 2. In VV ECMO, it is acceptable practice to target low PaO₂ and SpO₂ goals, sometimes as low as SpO₂ of 80%. This does not hold true in VA ECMO, and specifically in ECPR patients, in which hypoxia has been convincingly associated with poor neurological outcomes.
- g. Immediate post-cannulation initial blood pressure target for VA-ECMO, and/or any ECMO configuration with an arterial return cannula
 - i. Post Cardiectomy ECMO – goal MAP 60-80
 - ii. ECPR – goal MAP 65-75
 - 1. While post-cardiotomy ECMO patients are often managed with MAP 60-80, a MAP < 65 in the ECPR patient population, particularly, has been associated with poor neurologic outcomes.¹⁹⁻²¹ Similarly, MAP > 75 has also been associated with poor outcomes, though causality remains unclear
 - 2. Post-cannulation, if ECMO flow is within goal range (~2.4L/BSA/minute), any procedure-related bleeding is controlled,

and pulse pressure is > 15 , then the maximum blood pressure target should rapidly be liberalized to allow for permissive hypertension and optimal cerebral perfusion

- a. If liberalization of blood pressure leads to compromise of either ECMO flow, return cannula post membrane pressure, pulse pressure, or hemostasis, then a risk/benefit analysis must be made at the discretion of the ERS or CCM attending physician regarding the target blood pressure goal, optimizing both cerebral perfusion, and ECMO goals and their respective resuscitation end-points, as best as possible
 - b. While liberalizing the top BP target is reasonable and may be frequently indicated, lowering the BP floor target of MAP > 65 may be dangerous, and should not occur except in extreme circumstances
 - c. It would be highly unusual for a patient on VA-ECMO for ECPR to require antihypertensive medications. This perceived requirement should initiate a comprehensive review of the case, vasoactive medications, and the need for mechanical support
3. VA-ECMO, other ECMO configurations, without ECPR - Goal MAP 65-75
- a. Liberalization of blood pressure maxima is recommended once safe, provided it does not result in intravascular hemolysis, or compromise of ECMO flow, pulse pressure, return cannula post membrane pressures, or hemostasis
 - b. While most VA-ECMO patients should have similar requirements as post-ECPR VA-ECMO, these requirements are likely more flexible, particularly if there was never a primary brain injury or secondary brain injury, and the patient is neurologically intact with a reliable neurologic exam

B. Upon Arrival to ICU

a. Attach bilateral cerebral NIRS

i. Establish baseline RSO₂

1. If baseline RSO₂ < 60 , perform the following adjustments in sequence. Once RSO₂ reaches > 60 , no further adjustments are needed

- i. Note: any of the below steps can be omitted if viewed as dangerous or contraindicated by the CCM or cannulating team
- b. increase MAP goal to target 70-80mmHg
- c. If PaCO₂ < 40, decrease sweep gas to target goal PaCO₂ 40-50
- d. If either extremity arterial line pulse pressure is < 15, start an inotrope. Appropriate inotropes, and their respective dose ranges, are as follows:
 - i. Epinephrine 0.01 – 0.07 mcg/kg/min
 - ii. Dobutamine 2.5-10 mcg/kg/min
 - iii. Milrinone 0.125 – 0.75 mcg/kg/min
- e. Transfuse to goal Hb > 9-10
 - i. If already at Hb 9-10, transfuse 1U PRBC
 - ii. If Hb > 10, transfuse only at the discretion of the cannulating or CCM Team
- f. If there are continued signs of tissue hypoperfusion or shock, as potentially evidenced by persistent severe lactic acidosis, unchanged or worsening vasopressor requirements, CI or Fick calculation < 2.2, or capillary refill time > 3 seconds, ECMO flow may need to be increased
 - i. Please discuss with the ERS or CCM attending prior to increasing ECMO flow
 - ii. Consider an LV venting strategy, particularly if resuscitative endpoints are not properly downtrending as above, and pulse pressure is low despite inotropic support and the above measures described
- g. If all of the above interventions fail to improve RSO₂, no further interventions are indicated

b. Start continuous EEG

- i. EEG can be discontinued once the patient has a reliable neurologic exam and is consistently following commands, which, for the sake of this protocol, is described above
- ii. If the patient does not have a return of a reliable and reassuring neurologic exam, EEG can be discontinued at the discretion of the ERS and/or neurocritical care teams, when deemed no longer necessary for affecting management decisions

- c. **Transcranial Doppler exams** – obtain daily for the first 5 days post ECMO cannulation
- d. **Screen for and identify differential gas exchange**
 - i. Trend RUE ABG q4h
 - 1. If PP > 15 – titrate vent to goal PaCO₂ 40-45 and PaO₂ 80-140
 - 2. If PP < 15 – titrate ventilator AND FSO₂/sweep gas to target PaCO₂ 40-45 and PaO₂ 80-140
 - ii. Trend LUE Pulse Pressure q4h
 - 1. If LUE PP < 15 trend LUE ABG q4h in addition to standard blood gas monitoring
 - a. Place a LUE arterial line
 - b. Titrate vent to RUE goal PaCO₂ 40-45 and PaO₂ 80-140
 - c. Titrate FSO₂/sweep gas to target LUE goal PaCO₂ 40-45 and PaO₂ 80-140
- e. **Arterial blood gases** – while it is standard that the patient will have bilateral upper extremity arterial lines in this protocol, it is not recommended to draw routine arterial blood gases from both sites, an exception being in the circumstance of differential gas exchange. Arterial blood gases, as a rule, should only be drawn to answer a clinical question related to a specific clinical finding. Likewise, the following event protocols will outline when drawing bilateral blood gases has clinical merit.
 - i. When it is determined that bilateral arterial lines are no longer needed, such as when the patient has cardiac recovery, has a persistent return of a reliable and reassuring neurologic exam, and when ECMO is being weaned, the LUE arterial line should be discontinued in a timely fashion
- f. Daily reassessment and multidisciplinary discussion for consideration for discontinuation of the neuromonitoring protocol in clinically improved patients

EVENT PROTOCOLS

- A. **Definitions** – events are divided into two subcategories: Global hypoperfusion crisis and Regional hypoperfusion crisis (see below for formal definitions). These crises represent potential neurologic events and/or development of neurologic complications that may benefit from intervention and may require emergent neuroimaging
- B. **Global Hypoperfusion Crisis** – generally represents a global event, such as cerebral edema, brain herniation, or massive intracerebral hemorrhage. Global changes can also occur in seizures and status epilepticus. Finally, any sudden changes in cerebral blood flow of oxygenated blood can result in decreased NIRS, such as accidental disconnection or sweep gas, sudden change in gas exchange (i.e. sudden hypocapnia or hypoxemia)
- a. Definition: Any of the following
 - i. NIRS < 40%
 - ii. NIRS < 75% established baseline RSO₂
 - iii. Bilateral pupillometry < 3
 - b. Evaluation and management
 - i. Inspect the ECMO circuit for any changes
 - 1. Ensure the integrity and overall function of the ECMO circuit with the ECMO specialist
 - 2. Ensure the connection and function of the sweep gas
 - a. As a reminder, do not decrease F_sO₂ < 0.5LPM in VA ECMO patients, or in patients with an arterial return cannula
 - 3. If pulse pressure is <15, and/or has suddenly decreased, consider immediately decreasing ECMO flow to facilitate AV opening and LV unloading
 - ii. Repeat vital signs
 - iii. Repeat neurologic exam if safe and feasible
 - 1. If the patient has a reliable neurologic exam (See definition above), and it is reassuring and unchanged, interventions may be deferred per the attending physician's discretion
 - iv. Evaluate for change in pulse pressure on LUE and RUE arterial line waveforms
 - 1. Intervention
 - a. If pulse pressure > 20mm Hg or is unchanged from baseline, no intervention needed

- b. If pulse pressure has decreased from baseline, the following interventions can be considered
 - i. Bedside TTE to evaluate for aortic valve opening can assist with clinical decision-making
 - ii. Start inotrope or increase previously infusing inotropes. Preferred agents are listed per potential indications
 - 1. Hypertensive patients – Milrinone 0.125-0.75 mcg/kg/min
 - 2. Hypotensive patients – epinephrine 0.02-0.07 mcg/kg/min
 - 3. Normotensive patients – either agent may be appropriate, depending on the rest of the clinical scenario
 - iii. If lactate has already cleared, and vasopressors are off, consider decreasing ECMO flow
 - iv. Consider evaluation for LV vent, if indicated
 - 1. Cardiology consultation for Impella CP or intraaortic balloon pump
 - 2. Cardiac surgery consultation for Impella 5.5, atrial septostomy, or other surgical LV venting strategy
- v. Repeat blood gases from both RUE and LUE to evaluate for hypoxemia and hypocapnia
 - 1. Interventions
 - a. If RUE and LUE ABGs are equal
 - i. If RUE PP < 15, titrate F_{SO}2/sweep to target goal PaCO₂ >40 and PaO₂ < 80-140
 - ii. If RUE PP > 15, titrate vent to target goal PaCO₂ > 40 and PaO₂ < 80-140
 - 1. If patient-triggered ventilations are tachypneic and driving hypocapnia, consider increasing sedation to decrease respiratory drive
 - b. If RUE and LUE ABGs NOT equal
 - i. Titrate vent to target RUE parameters, and F_{SO}2/sweep to target LUE parameters

vi. Evaluate bilateral NIRS

1. If RSO₂ < 60 OR <75% established baseline RSO₂ (See above), perform the following interventions in sequence.
 - i. Once RSO₂ reaches either > 60 OR reaches previously established baseline RSO₂ (See above), no further adjustments are needed
 - ii. Note: any of the below steps can be omitted if viewed as dangerous or contraindicated by the CCM or cannulating team
 - b. increase MAP goal to target 70-80mmHg
 - c. If PaCO₂ < 40, decrease sweep gas to target goal PaCO₂ 40-50
 - d. If either extremity arterial line pulse pressure is < 15, start either epinephrine 0.01 – 0.07 mcg/kg/min, dobutamine 2.5-10 mcg/kg/min, or milrinone 0.125 – 0.75 mcg/kg/min
 - e. Transfuse to goal Hb > 9-10
 - i. If already at Hb 9-10, transfuse 1U PRBC
 - ii. If Hb > 10, transfuse only at the discretion of the cannulating or CCM Team
 - f. If there are continued signs of tissue hypoperfusion or shock, as potentially evidenced by persistent severe lactic acidosis, unchanged or worsening vasopressor requirements, CI or Fick calculation < 2.2, or capillary refill time > 3 seconds, ECMO flow may need to be increased
 - i. Please discuss with the ERS or CCM attending prior to increasing ECMO flow
 - ii. Consider placement of Impella for LV venting, particularly if resuscitative endpoints are not properly down-trending as above, and pulse pressure is low despite inotropic support and the above measures described
 - g. If all of the above interventions fail to improve RSO₂, no further NIRS-guided interventions are indicated
- vii. Consider repeat CT brain after stabilization, or after all stabilization options have been exhausted
- viii. If available, discuss with the EEG physician any acute EEG changes
- ix. Continue to trend pupillometry and NIRS, but consider deferring full neurologic exams and sedation holidays temporarily, until cleared by the

attending physician from either the ERS, CCM, or neuromedicine CCM teams

1. Increase analgo-sedation regimen to target a goal RASS < -2
 - a. If on TTM, consider a paralytic infusion to treat shivering

C. **Regional Hypoperfusion Crisis** – the main concern during a regional crisis is acute ischemic stroke or intracerebral hemorrhage, and thus, neuroimaging will often be required. However, the differential diagnosis also includes differential gas exchange, specifically when the mixing point is located in the aortic arch, between the take off of the carotid arteries. In such a situation, manipulation of ECMO (see below) and ventilator parameters may correct the regional crisis rapidly, and potentially obviate the need for neurologic imaging

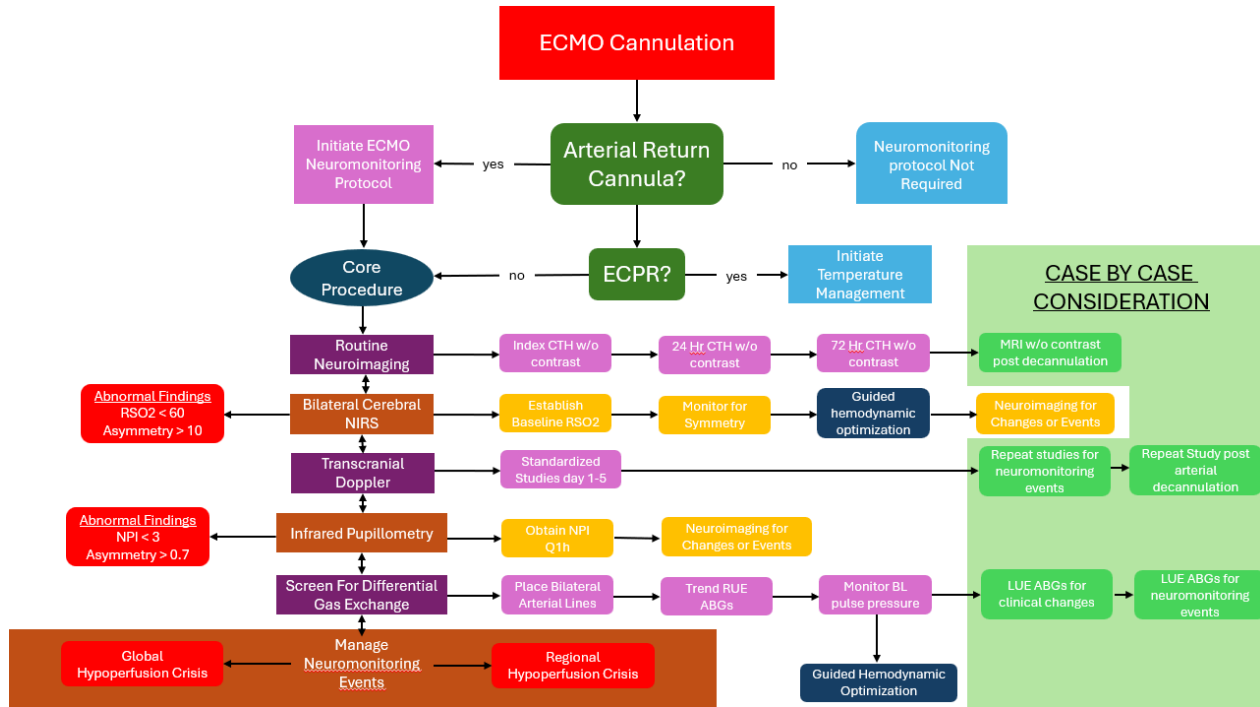
- a. Definition – any of the following
 - i. Single NIRS desaturation < 75% baseline RsO₂
 - ii. Single NIRS desaturation < 40%
 - iii. NIRS asymmetry < 15%
 - iv. Asymmetric NPI < 3 on pupillometry
 - v. NPI difference > 0.7% on pupillometry
- b. Evaluation and management
 - i. Immediate evaluation of ECMO
 1. Ensure the integrity and overall function of the ECMO circuit with the ECMO specialist
 2. Ensure the connection and function of the sweep gas
 - a. As a reminder, do not decrease F_sO₂ < 0.5LPM in VA ECMO patients, or in patients with an arterial return cannula
 3. If pulse pressure is < 15, and/or has suddenly decreased, consider immediately decreasing ECMO flow to facilitate AV opening and LV unloading
 - ii. Immediate brief evaluation of the ventilator
 1. Ensure the function of the ventilator and the connection to oxygen
 2. Assess and optimize usual ventilator parameters, such as tidal volume, positive end expiratory pressure (PEEP), and FiO₂.
 - a. Note: “Lung Rest” ventilator settings should never be utilized in patients on peripheral VA ECMO
 - iii. Immediate repeat comprehensive neurologic exam, unless contraindicated

1. If the patient has a reliable neurologic exam (See definition above), and it is reassuring, further interventions may be deferred per the attending physician's discretion
2. If a focal deficit is elicited on neurologic exam
 - a. Stat CTA head and neck perfusion scan
 - i. Note: in order to obtain an accurate CT head perfusion, an attending physician (either ERS, or CCM) will need to be present in the CT scanner to manage ECMO, which may involve either decreasing drainage or clamping the drainage cannula temporarily while contrast is administered
 - b. Consider holding heparin, if deemed safe from an LV unloading perspective
 - i. Note that just being on ECMO is not a contraindication to holding therapeutic anticoagulation
 - c. Consider initiating a stroke alert
 - i. Recommended for all patients with a new focal deficit on neurologic exam
3. If unable to obtain a reliable neurologic exam
 - a. Stat CTA head and neck perfusion. Consider a stroke alert if imaging findings are concerning for acute stroke or large vessel occlusion
 - i. Note: in order to obtain an accurate CT head perfusion, an attending physician (either ERS, or CCM) will need to be present in the CT scanner to manage ECMO, which may involve either decreasing drainage or briefly clamping the drainage cannula temporarily while contrast is administered. This is important to ensure contrast enters the cerebral circulation as intended for accurate imaging
- iv. Pupillometry
 1. If NPI asymmetry is present, assess MAR for recent albuterol administration or mydriatic agent administration
- v. Assess for differential gas exchange
 1. Evaluate LUE pulse pressure
 - a. Interventions: If pulse pressure < 15

- i. Consider adding an inotrope, or increasing inotrope infusion
 - ii. If lactate cleared, consider decreasing ECMO flow as an attempt to reestablish pulsatility in Lt CCA
- 2. Repeat RUE and LUE blood gases
 - a. If RUE and LUE ABGs NOT equal
 - i. Titrate vent to target RUE parameters, and FSO2/sweep to target LUE parameters
- 3. If RUE and LUE ABGs are equal, differential gas exchange is not likely the etiology of the focal crisis
 - a. Neuroimaging as above
 - b. Consider urgent repeat TCDs in AM to assess cerebral blood flow
 - c. Consider holding heparin temporarily until diagnostics are completed. This would be a risk/benefit analysis, and require coordination with ERS attending

APPENDIX

APPENDIX A – Protocol Summary



Neurologic Motor Exam

Respiratory and Cardiac components of SOFA score

Diagram of differential gas exchange

APPENDIX B - Limitations of neuroimaging

- Show image
- Procedure to obtain accurate perfusion imaging

KEYWORDS:

Extracorporeal cardiopulmonary resuscitation, ECMO, differential gas exchange, neurologic complications, and acute brain injury

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